A Treatise on Atomic Force Microscopy

Samriddha Ganguly 22279*
Department of Physics, IISER Bhopal

(Dated: April 1, 2025)

Abstract

In this term paper we will go about explaining the development of Atomic Force Microscopy (AFM), its theoretical background and need for the method. Further, we will try to decipher useful information from our own survey of Graphite sample under AFM while explaining the image processing and noise removal techniques employed by us.

I. BACKGROUND OF ATOMIC FORCE MICROSCOPY:

Atomic force microscopy (AFM) was developed when people tried to extend Scanning Tunneling Microscopy (STM) technique to investigate the electrically non-conductive materials, like proteins. In 1986, Binnig and Quate demonstrated for the first time the ideas of AFM, which used an ultra-small probe tip at the end of a cantilever [1]. In 1987, Wickramsinghe et al. developed an AFM setup with a vibrating cantilever technique [2], which used the light-lever mechanism. An AFM is rather

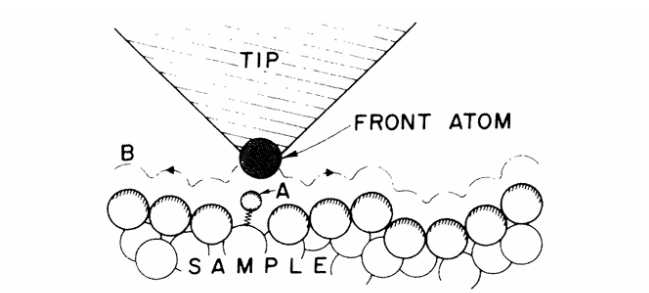

FIG. 1. Description of the principle operation of an STM as well as that of an AFM. The tip follows contour *B*, in one case to keep the tunneling current constant (STM) and in the other to maintain constant force between tip and sample (AFM, sample, and tip either insulating or conducting). The STM itself may probe forces when a periodic force on the adatom *A* varies its position in the gap and modulates the tunneling current in the STM. The force can come from an ac voltage on the tip, or from an externally applied magnetic field for adatoms with a magnetic moment.

FIG. 1. The description of AFM By Binning and Quate

different from other microscopes, because it does not form an image by focusing light or electrons onto a surface, like an optical or electron microscope. An AFM physically 'feels' the sample's surface with a sharp probe, building up a map of the height of the sample's surface. This is very different from an imaging microscope, which measures a two-dimensional projection of a sample's surface. Such a two-dimensional image does not have any height information in it, so with a traditional microscope, we must infer such information from the image or rotate the sample to see feature heights. The data from an AFM must be treated to form an image of the sort we expect to see from a microscope. This sounds like a disadvantage, but the treatment is rather simple, and furthermore it's very flexible, as having collected AFM height data we can generate images which look at the sample from any conceivable angle with simple analysis software. Moreover, the height data makes it very simple to quickly measure the height, length, width or volume of any feature in the image.

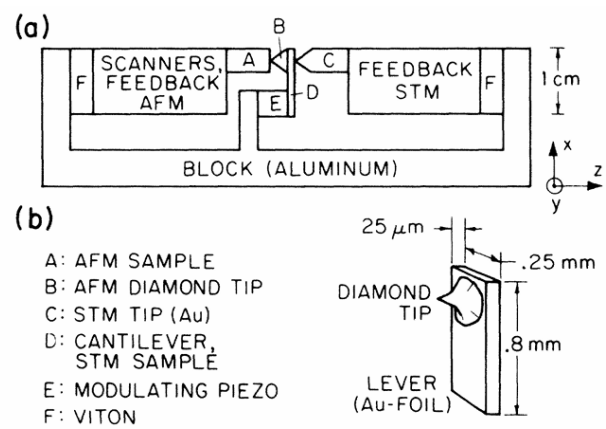

FIG. 2. Experimental setup. The lever is not to scale in (a). Its dimensions are given in (b). The STM and AFM piezoelectric drives are facing each other, sandwiching the diamond tip that is glued to the lever.

FIG. 2. Sketch of a Prototype AFM by Binning and Quate

A. Theoretical Background

AFM is a type of Near-Field Scanning Probe Microscopy. As mentioned before, the key difference which is one of the motivations for AFM applications is looking beyond the Diffraction Limit. AFM overcomes this by using various Tip-Surface interactions to physically image the surface. Depending on the situation, forces that are measured in AFM include mechanical contact force, van der Waals forces, capillary forces, chemical bonding,

* Contact author: samriddha22@iiserb.ac.in

electrostatic forces, magnetic forces (see magnetic force microscope, MFM), Casimir forces, solvation forces, etc.

1. *The Lennard-Jones Potential*

We have learnt about the Van der Waal's Attractive forces and the Pauli Repulsion forces which act in different regimes and how that leads to the intermolecular pair potential known as the **Lennard-Jones Potential**. When we consider the London Dispersion Forces and experimental data taken by Lennard-Jones in 1931, we obtain the following mathematical form of the 12-6 Potential:

$$V_{\mathrm{LJ}}(r) = 4\epsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right] \tag{1}$$

Where:

- V_{LJ} is the potential energy
- r is the distance between particles
- ϵ is the depth of the potential well
- σ is the finite distance where potential is zero

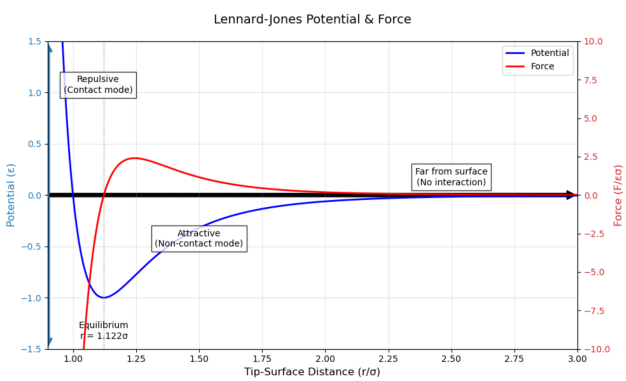

FIG. 3. Lennard-Jones Potential & Force showing the attractive (non-contact) and repulsive (contact) regimes, with equilibrium at $r = 1.122\sigma$.

We have plotted the same along with its force and we will soon show how different modes of AFM exploit different regions of the plot as mentioned here.

2. *The Working Principle*

An atomic force microscope (AFM) works by scanning a tiny, flexible probe with an ultra-sharp tip across a sample surface. As the probe moves in a back-and-forth pattern, the tip interacts with the surface, causing the cantilever to bend up and down as it encounters bumps and dips.

A laser beam reflects off the cantilever onto a sensitive light detector that tracks these subtle movements - think of it like watching a flashlight beam wiggle on a wall as the mirror moves. The detector converts these light position changes into electrical signals that scientists calibrate to determine exactly how much the tip moved in nanometers.

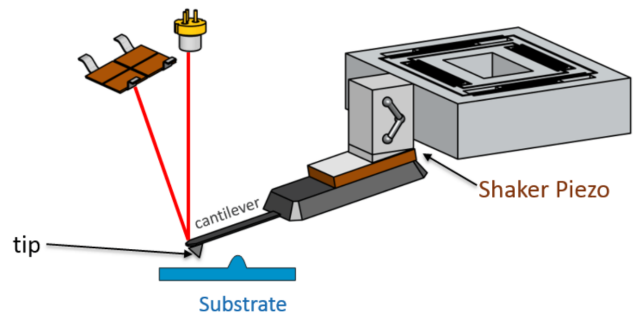

FIG. 4. Schematic of Working of Components of AFM

By carefully measuring these deflections at every point across the sample, the AFM builds a detailed 3D map of the surface, revealing features as small as individual atoms. This all happens through precise coordination between the probe's motion, laser detection system and computer analysis which will be explained in the latter sections.

B. Instrumentation and Modes of AFM

AFM is an assembly of a number of important parts which perform pivotal functions leading upto the imaging of the substrate. These can be categorized as follows:

- Cantilever/AFM Tip Assembly
- Laser Diode and Position-Sensitive Photo-detector System (Optical Lever Sensor)
- Feedback Loop
- Piezoelectric Scanner
- Processing Software and Machine (Typically a Computer)

Here we have used the **NanoSurf EasyScan 2 Controller** Machine along with the NanoSurf Software and the AFM model is called **NanoSurf easyScan 2 AFM**. We will delve deeper into the nuances of each instrument and explain their functionality.[3]

1. *Cantilever-Tip Assembly*

This assembly consists of a very sharp tip (typical radius of curvature at the end for commercial tips is 5-10

nm) that hangs off the bottom of a long and narrow cantilever. As mentioned previously, the cantilever/tip assembly is also referred to as the AFM probe. The length-/height of the AFM cantilever tip varies depending on the type of cantilever. The two most common geometries for AFM cantilevers are rectangular ("diving-board") and triangular.

AFM cantilever material typically consists of either silicon or silicon nitride, where silicon nitride is reserved for softer cantilevers with lower spring constants. The dimensions of the cantilever are very important as they dictate its spring constant or stiffness. This stiffness is fundamental to governing the interaction between the AFM cantilever tip and the sample surface and can result in poor image quality if not chosen carefully. The relationship between the cantilever's dimensions and spring constant, k, is defined by the equation:

$$k = \frac{Ewt^3}{4l^3} \tag{2}$$

where

- w=cantilever width
- t=cantilever thickness
- L=Cantilever Length
- E= Young's Modulus of the Material

2. *Optical Lever System*

A laser beam is reflected by the back side of a reflective cantilever onto a four-segment photodetector. If a probe, mounted on the front side of the cantilever, interacts with the surface the reflected light path will change. The force is then measured by monitoring the change in light detected by the four quadrants of the photodetector and the Topography is plotted accordingly by the measurement software.

3. *Raster Scanning*

AFM Follows a Raster Scanning pattern. It can be visualized quite easily using 5. Basically in Raster scan we sweep horizontal lines with certain number of points which determine the step size. The parameter that can be varied here are (*i*) Time per line (*ii*) Points per line (Points where it scans (*iii*) Number of lines. We see that increasing all these in a 2D raster scan would in fact lead us to better resolution of images obtained from our AFM. Now we shall know how this scanning is carried out.

4. *Feedback Loop*

Piezoelectric materials are electromechanical transducers that convert electrical potential into mechanical motion. In AFM, they are essential to our Feedback Control Mechanism.

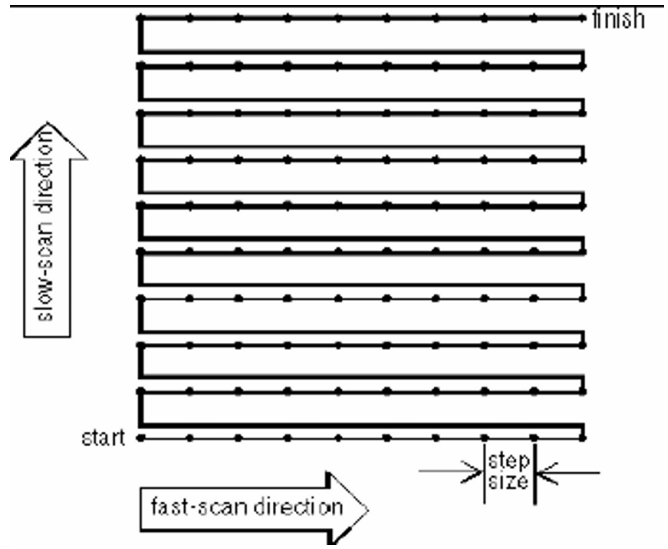

FIG. 5. Schematic of Working of Components of AFM

The force between an AFM probe and a surface is measured with a force transducer. When the probe comes into contact with the surface, the voltage output from the transducer increases. It is important that the output of the transducer be monotonic and increases as a greater force is applied between the probe and surface. Force transducers may be constructed that measure forces as low as 10 piconewtons between a probe and a surface. Typically, the force transducer in an AFM is a cantilever with integrated tip (the probe), and an optical lever; however, there are several types of force sensors that may be used in an AFM. Some of these are as follows:

- **Scanning Tunnelling Microscope:** In the original AFM built in 1985 a scanning tunneling microscope tip was use to measure the motion of a cantilever. Although this technique was viable, implementation and operation were very difficult.
- **Crystal oscillator:** A piezoelectric crystal such as quartz can be used to measure the force between a probe and a surface. If the probe mounted on the crystal is vibrated and positioned close to a surface, the interaction of the probe and surface will cause a change in the vibration. This change is proportional to force.
- **Piezo-resistive cantilevers:** A cantilever can be fabricated that has a small piezo-resistive element in it that changes resistance if the cantilever bends.

Essentially, the feedback loop is what "runs" the scanning. In AFMs, depending on the different modes, there is a parameter that serves as the setpoint. For example, in static mode (contact mode) the feedback parameter is cantilever deflection, while in the most common form of

tapping mode, the cantilever oscillation amplitude is the feedback parameter. The instrument is trying to keep this feedback parameter constant at its setpoint value by adjusting the z-piezo to move the cantilever probe up and down. The resulting z-piezo movements provide the height information to create the surface topography.

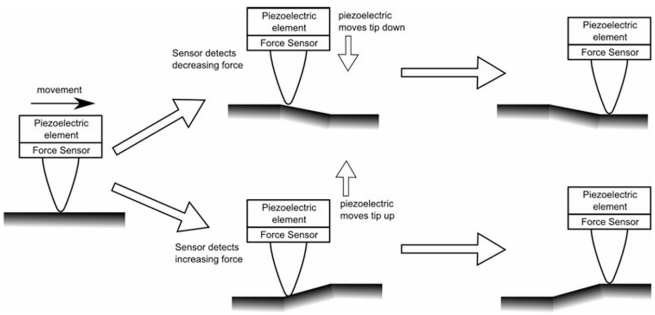

FIG. 6. Schematic of Working of Components of AFM[4]

As illustrated in Figure 6, the control electronics take the signal from the force transducers, and use it to drive the piezoelectric (particularly the Z-piezo) so as to maintain the probe–sample distance, and thus the interaction force at a set level. Thus, if the probe registers an increase in force (for instance, while scanning the tip encounters a particle on the surface), the feedback control causes the piezoelectric to move the probe away from the surface. Conversely, if the force transducer registers a decrease in force, the probe is moved towards the surface.

5. *x-y-z Piezoelectric Scanners*

The most common types of piezoelectric materials in use for AFM scanners are constructed from amorphous lead barium titanate, $PbBaTiO_3$norlead zirconate titanate, $Pb[Zr_xTi_1{-}x]O_3$, $0 < x < 1$. Electromagnetic scanners provide highly accurate and precise nanoscale motion in X, Y, and Z at low operation voltage in Nanosurf AFMs. These kinds of scanners provide significant advantages of highly linear motion and the absence of creep over other kinds of scanners such as piezoelectric scanners. The Nanosurf FlexAFM-based systems combine a piezoelectric scanner for Z motion with a flexure-based electromagnetic scanner in X and Y; this configuration provides fast motion in Z for raster scanning with maximum flatness in X and Y, which is optimal for the advanced capabilities offered by these systems.

In order to develop an AFM that is able to scan much faster than normal, the scanner must be able to overcome the limitation of the traditional scanners, which is their low first resonant frequency. A scanner with a higher resonant frequency will allow faster scanning without the scanner going into resonance.

Ando et al. have made significant progress in this direction. For example, a fast scanner has been constructed from piezoelec tric stacks, to achieve a high resonant frequency of 240 kHz versus 15 kHz for a typical tube scanner. [5]

Now we will give a schematic of how we can arrange these scanners and the Optical Lever System to form a working AFM as given in figure 7

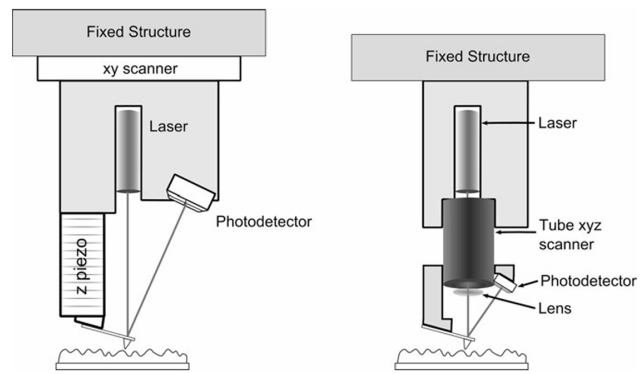

FIG. 7. Designs for tip-scanning AFMS with optical lever sensors. Left: the laser is scanned with the cantilever. Right: the laser is fixed and the cantilever is scanned, a lens keeps the laser light focused on the cantilever.

Thus we have successfully assembled a working AFM!

C. Modes in AFM

AFM can be utilized in different modes for both imaging as well as force calculation purposes. Some of these are given below.

1. *Static AFM Modi*

Contact mode: Tip in continuous contact with sample preferably used for hard samples imaging in air and liquid. High resolution is obtained. The deflection of cantilever is kept constant and it is measured. Contact mode imaging is heavily influenced by frictional and adhesive forces, and can damage samples and distort image data.

Force Spectroscopy Mode: Consecutive cycles of tip approach and retract. Interaction forces between tip and sample are recorded.

2. *Dynamic AFM Modi*

Intermittent/tapping mode: Oscillating cantilever, tip touching surface gently and frequently. Often used for biological samples. Imaging in air and liquid with good resolution.

Non contact mode: Oscillating cantilever, tip not in contact with sample. Used for soft samples. Imaging in vacuum. It does not damage the tip surface or tampering

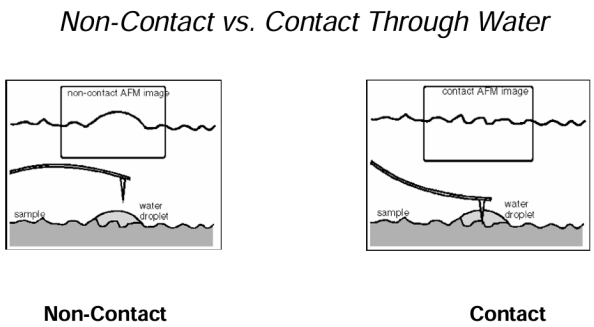

FIG. 8. Advantage of Contact Mode Scanning

with the sample. We have used this here. However an unique disadvantage with the contact mode is shown in Figure 8.

D. Notes

We may also find the correlation of tip dimensions with the signal to noise ratio using the equipartition theorem to characterize the thermal noise for a particular spring constant k_S, given by $k_S < \Delta x^2 >= k_B T$. We thus see the smaller the length of the tip less is the mean square displacement due to the Thermal Noise. Thus smaller length ensures a better signal-to-noise ratio.[6][7]

Further we may go on to state some limitations of AFM:

1. **Tip-Sample Artifacts:** Tip convolution effect distorts images (tip shape affects measured features). Sample deformation from excessive tip force (especially soft samples). Tip wear during scanning alters resolution
2. **Maximum practical scan area:** typically $< 100x100$ m. Large-area scans suffer from thermal drift and piezo nonlinearities.
3. **Environmental Sensitivity:** Vibration/noise isolation required (sensitive to acoustic/mechanical noise). Thermal drift affects long measurements (room temperature fluctuations). Air humidity can influence measurements.

II. EXPERIMENT & ANALYSIS

A. Procedure and Use of Software:

I will now list down the steps followed by us to perform the experiment.

1. We first connect the AFM module with the NanoSurf Easyscan Controller to connect the machine to the Software.
2. First position the cantilever over the sample that you want image.
3. Now to image using the Non-Contact Mode the tip must come within a fraction of a nanometer to the sample
4. We initially have a manual course approach named as Advance and then approach it finely after a certain depth by eye-check using the Approach button.
5. the scan by setting the parameters namely image size, time/line and Points per line.
6. After scan is completed we can retract the tip using the retract button and then manually coarsely remove it using the Withdraw button.
7. Using this we have thus obtained the AFM image of a given sample and we may move forward towards image analysis techniques.

B. Image Analysis and Characterization of Sample

Processing steps change the AFM data; they include functions like filtering and back ground subtraction. Occasionally an image may not need any processing at all, but this is not common with most samples. All processing is done with the aim of clarifying the data already within the file. The purpose is to make it easier to measure and characterize as we please. We will go into some of the main elements of this processing and then delve into our experimental findings.

1. Levelling

The reason levelling is required is that AFM images usually measure sample height. If the background in the image (such as the substrate on which the sample was deposited) has considerable tilt in it, the change in height of the background will mask the changes in height associated with the sample. AFM is often used to measure samples with very small heights, so even a small tilt in the sample background can have serious effects.

In addition to tilting in the image caused by the AFM and sample not being perfectly orthogonal, a common problem in AFM images is scanner bow. It occurs mainly in instruments that use tube scanners, and is caused by a swinging motion of the free end of the scanner. This leads to a curve in the image plane. There are a number of different methods that can be used for image levelling. These include the following:

- Polynomial Fitting: A very common method for levelling AFM images is by polynomial fitting, or 'line-by line' levelling. In this routine, each line in the image is fit to a polynomial equation. Then, the polynomial shape is subtracted from the scan line,

which leads to the line being not only flattened, but also shifted such that it centres on zero height.

- Two Dimensional Plane Fitting: (We Have used this method in our codes) This technique attempts to independently warp a flat plane to the image and then subtracts the best fit. aircraft; it performs well under conditions with a completely flat terrain, and makes no use of Curvature. As a default, the process will automatically take into account the whole image for fitting; this This means that it could be susceptible to inaccuracies since the big topographic features will lower the accuracy of the background fit. Plane fitting does not result in the type of errors mentioned above which may afflict line-by-line fitting. It is therefore a fairly conservative levelling procedure, since although it is not particularly effective with lots of images, it doesn't introduce any errors.

2. Filtering

Atomic Force Microscopy (AFM) data often contains noise from mechanical vibrations, electronic interference, and thermal fluctuations. Filtering helps remove noise while preserving real surface features. It can be done using both Matrix and Fourier Filters. Matrix Filters are again classified as High-Pass and Low-Pass Filters. .Low-pass filters are so-called because they allow low-frequency components of the image to pass,while reducing the high-frequency components.This has a smoothing effect on the data,and so removes the high-frequency noise commonly found in AFM images.Over-application may blur data.High-pass filters,on the other hand,allow high-frequency components to pass while reducing low-frequency components.They are usually used for sharpening or edge detection in AFM images.

In Fourier Filtering we perform FFT and apply certain FFT based filters namely Low-Pass Filter (As we have used), Gaussian Filters, etc. Basically here the frequency separation is implemented in the Fourier/Frequency Domain. Here the Low-Pass filter is particularly useful as it has a two-fold advantage:

- Removes high-frequency noise (e.g., scanner vibrations, electronic noise).
- Preserves slowly varying surface features (e.g., large-scale topography).

Low pass filtering has its own pitfalls which can be stated as follows:

- Over-filtering blurs sharp features (e.g., step edges, nanoparticles).
- May mask true high-frequency details (e.g., atomic lattice defects).

3. Presentation of AFM Images

The Most basic representation possible for AFM images is **3-D rendering of levelled and filtered image**. AFM height data is inherently three dimensional (3-D). However, the standard method of rendering AFM data shows a two-dimensional (2-D) image, using a colour scale to represent height information. This is not a normal way for humans to see shapes, and can make interpretation difficult. In particular, for viewers unused to AFM data, it can be difficult to determine which features are higher than others, etc. One way to overcome this is to render the height information as a pseudo-three-dimensional image. This is easily achieved through conventional coding languages like Python by converting the colour gradient to a height in arbitrary units.

Another important method of representation of AFM data is the use of **Line profiles**. They help us to extract a lot of useful data from the measurements. Because AFM images are difficult to measure dimensions directly from, line profiles are usually extracted in order to measure dimensions from the AFM images. The AFM analysis software allows the user to arbitrarily define lines to be extracted, and these can be horizontal, vertical or at any angle. The software then constructs a new plot, with distance along the chosen line on the x axis, and sample height on the y axis. In the new plot, the user can easily measure feature heights, widths, and angles.

A non-visual property of a sample that may be extracted from AFM data is the **Roughness** of the surface. The most commonly used roughness parameters are probably the arithmetic roughness, (R_a) and the root-mean-squared roughness (R_q or R_{rms}). Both of these values have a positive correlation, i.e. larger values mean greater topographical variation in the image. Often, both values give rather similar results, but R_q will always be somewhat larger than R_a, and is rather more sensitive to outlying points than Ra. The formulas for these parameters along with some other commonly calculated roughness parameters are shown as follows:

$$R_a = \frac{1}{n}\sum_{i=1}^{n}|y_i| \tag{3}$$

$$R_q = \sqrt{\frac{1}{n}\sum_{i=1}^{n}y_i^2} \tag{4}$$

Roughness is a very important surface property for technological applications, but is also widely used as a quantitative measurement of surfaces changes in a wide range of samples. For example, absorption of materials to a surface, or erosion of a surface will typically lead to an increase in roughness. Another interesting venture is to do a $2-D$ **Autocorrelation Analysis.** Autocorrelation measures how similar a signal or image is to itself at

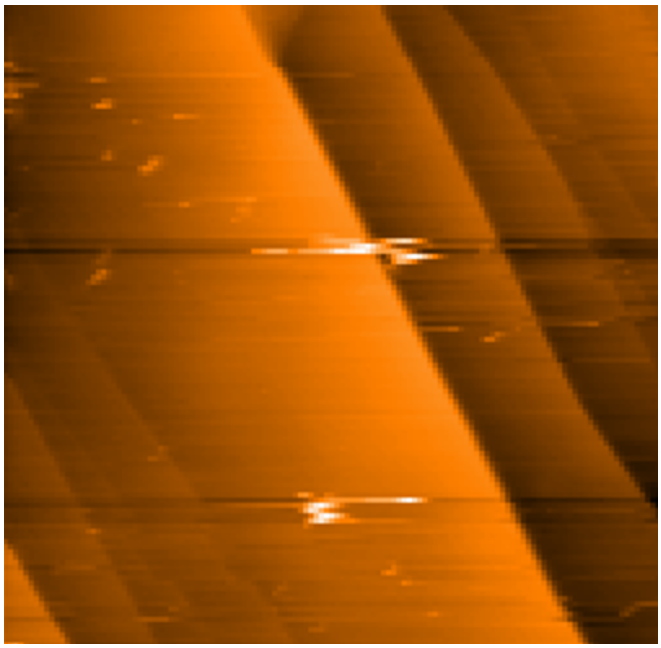

FIG. 9. Obtained Graphite Image. Parameters: 50x50 μm, Time/Line=1s, Points/Line=512

different spatial shifts (lags). In 2D autocorrelation analysis, we analyze the spatial periodicity and texture of an AFM (Atomic Force Microscopy) image by computing how each pixel correlates with its neighbors across the entire image.[4]

Interpretation of Autocorrelation:

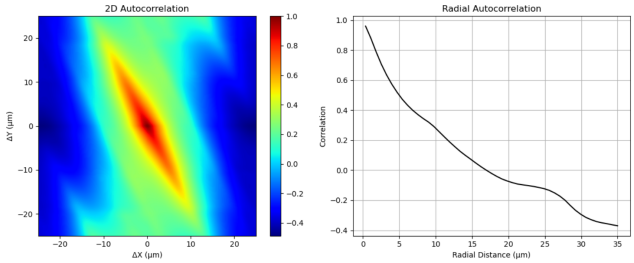

FIG. 10. 2-D Autocorrelation Analysis and the Radial Decay Profile

- Peak at (0,0): Maximum correlation (image compared to itself).
- Secondary peaks: Indicate repeating features (e.g., periodic nanostructures).
- Decay rate: Measures how quickly correlation is lost (related to surface randomness).
- Grain size analysis: Distance between peaks feature spacing.
- Surface anisotropy: Elliptical autocorrelation contours indicate directional trends.

C. Experimental Observations

We have arrived at the analysis of our experimental findings. We look at our obtained Graphite image using the AFM in figure 9.

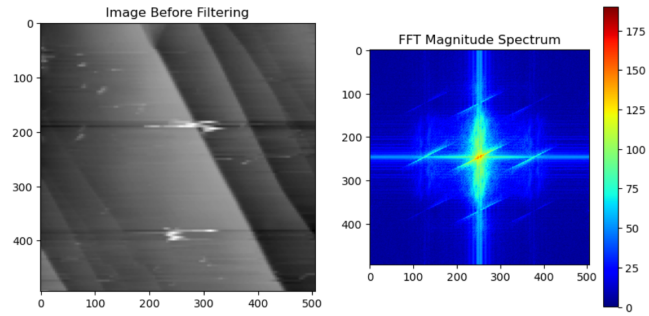

FIG. 11. Left: Grayscale Graphite Image Before Filtering. Right: FFT Spectrum

We move forward and convert this to grayscale and obtain the levelled image. Further, we use the FFT method and apply a low-pass filter to obtain the smoothened image. After the FFT Filtering is done and we have obtained the filtered image we can move forward and go to other characterization like the 3-D Rendering in Figure 13 and line profiles we have obtained in Figure 14.

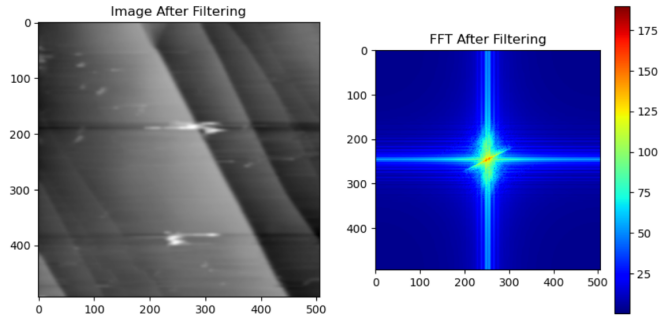

FIG. 12. Left: Grayscale Graphite Image After Filtering. Right: FFT Spectrum after Filtering

The Autocorrelation also shows the peak correlation near the centre and the decay profile shows the surface randomness as exhibited in figure 10.

III. CONCLUSION

We have thus characterized the Graphite surface and obtained the flake-like structure of the sample. We have also found out the roughness of the given region as $R_a = 0.0981$ units and $R_q = 0.1193$ in Z-axis. We have done line-profiling and Autocorrelation analysis (2-D) which results have been shared as figures. Atomic Force Microscopy in summarization is a truly remarkable innovation which has applications in varied domains

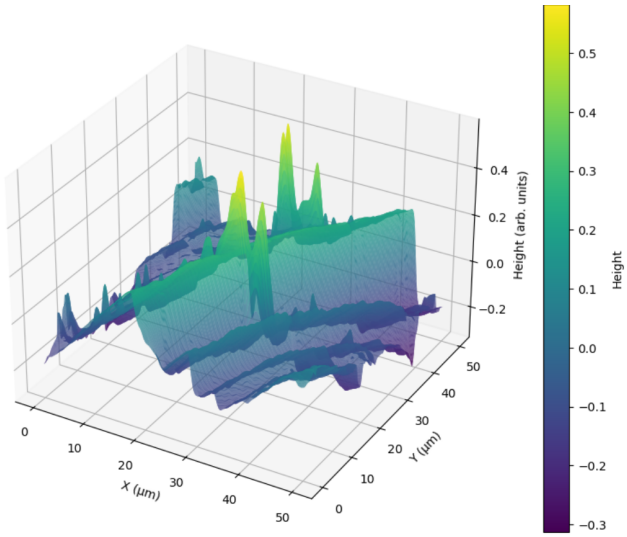

FIG. 13. 3-D Rendering of Final AFM Image of Graphite

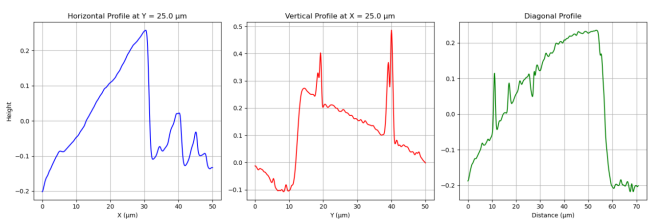

FIG. 14. Line Profiles of the obtained Graphite Image at $25\mu m$ along both X and Y coordinates and one along the diagonal

such as imaging of biological structures in air, 2-D crystals in liquid, nanolithography as it can interact with the sample, force spectroscopy and interpretation of force-distance curves, visualisation of domains, etc.

ACKNOWLEDGMENTS

I would like to acknowledge Adarsh's efforts in letting us have such a free-thinking oriented and exploratory lab course. I truly enjoyed and cherished each day I spent working on this. I would also like to thank my lab group members as well as my Physics Major friends who helped me in both performing and understanding the nuances of Atomic Force Microscopy.

Appendix A: Python Code for AFM Image Analysis

This presents a Python script for processing Atomic Force Microscopy (AFM) images, including leveling, filtering, 3-D visualization, and roughness analysis.

PYTHON IMPLEMENTATION

```python
import numpy as np
import matplotlib.pyplot as plt
from skimage import io, color, filters
from scipy import ndimage
from mpl_toolkits.mplot3d import Axes3D
from matplotlib import cm

# Load the AFM image
image_path = r"C:\Users\SAMRIDDHA\Pictures\Screenshots\Screenshot 2025-03-12 115110.png"
color_image = io.imread(image_path)

# Convert to grayscale if it's a color image
if len(color_image.shape) == 3 and color_image.shape[2] in [3, 4]:
    gray_image = color.rgb2gray(color_image[:, :, :3])  # Ignore alpha channel if present
else:
    gray_image = color_image  # Assume it's already grayscale

# Display original image
plt.figure(figsize=(10, 5))
plt.subplot(1, 2, 1)
plt.imshow(color_image)
plt.title('Original AFM Image')
plt.subplot(1, 2, 2)
plt.imshow(gray_image, cmap='gray')
plt.title('Grayscale Conversion')
plt.show()

## Step 1: Leveling (Background subtraction)
def level_image(image):
    # Fit a plane to the image and subtract it
    rows, cols = image.shape
    x, y = np.meshgrid(np.arange(cols), np.arange(rows))

    # Fit a plane using least squares
    A = np.column_stack((x.ravel(), y.ravel(), np.ones(rows*cols)))
    b = image.ravel()
    coeff, _, _, _ = np.linalg.lstsq(A, b, rcond=None)

    # Create the plane
    plane = coeff[0] * x + coeff[1] * y + coeff[2]

    # Subtract the plane
    leveled = image - plane
    return leveled

leveled_image = level_image(gray_image)

# Display leveled image
plt.figure(figsize=(10, 5))
plt.subplot(1, 2, 1)
plt.imshow(gray_image, cmap='gray')
plt.title('Original Grayscale')
plt.subplot(1, 2, 2)
plt.imshow(leveled_image, cmap='gray')
plt.title('After Leveling')
plt.show()
```

```python
## Step 2: Low-pass filtering with FFT visualization
def low_pass_filter(image, sigma=2):
    # Compute FFT
    fft_image = np.fft.fft2(image)
    fft_shifted = np.fft.fftshift(fft_image)
    magnitude_spectrum = 20 * np.log(np.abs(fft_shifted) + 1)  # Log scale for visualization

    # Display FFT before filtering
    plt.figure(figsize=(10, 5))
    plt.subplot(1, 2, 1)
    plt.imshow(image, cmap='gray')
    plt.title('Image Before Filtering')
    plt.subplot(1, 2, 2)
    plt.imshow(magnitude_spectrum, cmap='jet')
    plt.title('FFT Magnitude Spectrum')
    plt.colorbar()
    plt.show()

    # Apply Gaussian filter
    filtered = filters.gaussian(image, sigma=sigma)

    # Compute FFT after filtering
    fft_filtered = np.fft.fft2(filtered)
    fft_shifted_filtered = np.fft.fftshift(fft_filtered)
    magnitude_spectrum_filtered = 20 * np.log(np.abs(fft_shifted_filtered) + 1)

    # Display FFT after filtering
    plt.figure(figsize=(10, 5))
    plt.subplot(1, 2, 1)
    plt.imshow(filtered, cmap='gray')
    plt.title('Image After Filtering')
    plt.subplot(1, 2, 2)
    plt.imshow(magnitude_spectrum_filtered, cmap='jet')
    plt.title('FFT After Filtering')
    plt.colorbar()
    plt.show()

    return filtered

filtered_image = low_pass_filter(leveled_image)

## Step 3: Solid 3D Rendering
def render_3d(image, title='AFM 3D Surface', elevation=30, azimuth=45):
    rows, cols = image.shape
    x, y = np.meshgrid(np.arange(cols), np.arange(rows))

    fig = plt.figure(figsize=(12, 8))
    ax = fig.add_subplot(111, projection='3d')

    # Create solid surface plot with more vibrant colors
    surf = ax.plot_surface(x, y, image, cmap=cm.viridis,
                           linewidth=0, antialiased=True,
                           rstride=1, cstride=1,
                           alpha=1.0)  # No transparency

    # Adjust view angle
    ax.view_init(elev=elevation, azim=azimuth)

    ax.set_title(title, fontsize=14)
    ax.set_xlabel('X (pixels)', fontsize=12)
    ax.set_ylabel('Y (pixels)', fontsize=12)
    ax.set_zlabel('Height', fontsize=12)

    # Add color bar
    cbar = fig.colorbar(surf, shrink=0.6, aspect=10)
    cbar.set_label('Height', rotation=270, labelpad=15)

    plt.tight_layout()
    plt.show()

render_3d(filtered_image, 'Processed AFM Image', elevation=40, azimuth=30)

## Step 4: Roughness Analysis
def calculate_roughness(image):
    # Average roughness (Ra)
    Ra = np.mean(np.abs(image - np.mean(image)))

    # RMS roughness (Rq)
    Rq = np.sqrt(np.mean(np.square(image - np.mean(image))))

    return Ra, Rq

Ra, Rq = calculate_roughness(filtered_image)
print(f"Average Roughness (Ra): {Ra:.4f}")
print(f"RMS Roughness (Rq): {Rq:.4f}")

## Step 5: Line Profiles
def plot_line_profiles(image, num_profiles=3):
    rows, cols = image.shape

    plt.figure(figsize=(15, 5))

    for i in range(num_profiles):
        # Random positions
        if i == 0:
            # Center horizontal profile
            row = rows // 2
            col_start, col_end = 0, cols
            profile = image[row, :]
            title = f'Horizontal Line Profile (Row {row})'
        elif i == 1:
            # Center vertical profile
            col = cols // 2
            row_start, row_end = 0, rows
            profile = image[:, col]
            title = f'Vertical Line Profile (Column {col})'
        else:
            # Random diagonal profile
            if np.random.rand() > 0.5:
                # Top-left to bottom-right
                row_start, col_start = 0, 0
                row_end, col_end = rows-1, cols-1
            else:
                # Top-right to bottom-left
                row_start, col_start = 0, cols-1
                row_end, col_end = rows-1, 0
            profile = ndimage.map_coordinates(image,
                                              np.array([np.linspace(row_start, row_end, 1000)
```

```python
    ,
        np.linspace(col_start, col_end, 1000)]))
            title = 'Diagonal Line Profile'

        plt.subplot(1, num_profiles, i+1)
        plt.plot(profile, linewidth=2)
        plt.title(title, fontsize=12)
        plt.xlabel('Position', fontsize=10)
        plt.ylabel('Height', fontsize=10)
        plt.grid(True, alpha=0.3)

    plt.tight_layout()
    plt.show()

plot_line_profiles(filtered_image)

## Step 6: 2D Autocorrelation Analysis
def autocorrelation_analysis(image):
    # Subtract mean
    image = image - np.mean(image)

    # Compute 2D autocorrelation using FFT
    fft_img = np.fft.fft2(image)
    autocorr = np.fft.ifft2(fft_img * np.conj(fft_img)).real
    autocorr = np.fft.fftshift(autocorr)
    autocorr = autocorr / np.max(autocorr)  # Normalize

    # Plot autocorrelation
    plt.figure(figsize=(12, 5))
    plt.subplot(1, 2, 1)
    plt.imshow(autocorr, cmap='jet')
    plt.title('2D Autocorrelation', fontsize=12)
    plt.colorbar()

    # Radial profile
    center = np.array(autocorr.shape) // 2
    x, y = np.indices(autocorr.shape)
    r = np.sqrt((x - center[0])**2 + (y - center[1])**2)
    r = r.astype(int)

    radial_mean = np.bincount(r.ravel(), autocorr.ravel()) / np.bincount(r.ravel())

    plt.subplot(1, 2, 2)
    plt.plot(radial_mean[:min(autocorr.shape)//2], linewidth=2)
    plt.title('Radial Autocorrelation Profile', fontsize=12)
    plt.xlabel('Distance (pixels)', fontsize=10)
    plt.ylabel('Correlation', fontsize=10)
    plt.grid(True, alpha=0.3)
    plt.tight_layout()
    plt.show()

autocorrelation_analysis(filtered_image)
```

Listing 1. AFM Image Processing Script

CODE DESCRIPTION

The script performs the following operations:

- Image loading and grayscale conversion
- Background leveling by plane subtraction
- Low-pass filtering with FFT visualization
- 3D surface rendering
- Roughness quantification (Ra and Rq)
- Line profile extraction
- 2D autocorrelation analysis

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